

Quality control

Making the most of herbal medicine means ensuring that herbs and herbal products used are of good quality—properly grown, well dried, correctly processed, and within their sell-by date. Using poor-quality herbal produce is all too often a waste of money since there is the strong possibility that you will receive little benefit from it. When it comes to herbal medicine, quality is everything.



Quality is vital for herbal medicine. Without a guarantee that the correct herb of the right quality is being used, it is hard to be confident that the medicine will prove effective. In fact, one reason why the medical profession has generally preferred conventional medicines to herbal ones is the difficulty of guaranteeing quality in herbal remedies. Many herbal products on the market are of high quality but some can be very poor. A 2006 U.S. study of black cohosh (*Cimicifuga racemosa*, p. 79) found that only 7 out of 11 over-the-counter products tested contained what was stated on the label. Four contained a cheaper Chinese species instead. The adulteration of herbal products, particularly in Chinese and Indian herbs sold in the West is, regrettably, not unusual.

Herb quality may be affected not only by deliberate adulteration, but by the use of wrongly identified or poor-quality material. The herb may have been poorly harvested, dried, or stored, or it may be old or decayed. It may even be that the wrong herb was used. In each case, the lack of attention to quality results in a product with reduced medicinal value—or even none at all.

To try to ensure that only good-quality products are made, manufacturers of genuine herbal medicines use strict quality-

control procedures. Usually this involves comparing the dried herb material with listings in an herbal or national pharmacopeia (a standard reference work that gives the characteristics one would expect to find when analyzing a specific herb). Quality control involves making routine checks to establish that the herbal material is what it claims to be and that it meets a number of minimum requirements. The material is inspected with the naked eye, and assessed microscopically, to see if its botanical profile matches the standard. Other checks are made to see whether it contains appropriate levels of active constituents and to ensure that the material is free from contamination.

More sensitive quality-control methods, however, recognize that the quality of an herb does not depend simply on the presence of one or two key active constituents. Increasingly, people in the world of herbal medicine are focusing on the “fingerprint” of an herb—the unique chemical profile that represents the complex pattern of constituents found when good-quality dried herb material is analyzed by sensitive scientific machinery. By monitoring the sample and comparing it with this unique fingerprint, it is possible to make a much broader assessment of quality than when using only one or two constituents as a standard.

Buying Herbal Medicines

What should you look for when buying herbal medicines? It is usually most convenient to buy capsules, tablets, essential oils, pessaries, and perhaps tinctures, and to make up your own infusions, decoctions, and syrups (see p. 291 and p. 293).

- Buy from a reputable herb store, staffed by people knowledgeable about herbal medicines.
- Only buy herbs online from established herbal suppliers.
- Buy organic herbs and products where available.



Calendula

Buying Dried Herbs

Dried herbs are generally available from herbal suppliers. Buying from shops is preferable to buying online because the herbs can be examined before purchasing. However, it is possible some online companies may supply fresher herbs due to higher turnover. To gain the best medicinal effect, good-quality produce is essential. Shop around and bear in mind the following points before buying:

- Herbs should not be stored in clear glass jars or in direct sunlight, as this causes oxidation, which affects their efficacy.

- Good-quality aromatic herbs should have a distinct scent and taste.
- Check for signs of infestation due to poor drying techniques, or adulteration. This can sometimes be recognized by the presence of dried grass or other non-medicinal material in the jar.
- Herbs lose their color as they age. Look for bright material that has been well dried and stored, and that is not too old. Calendula flowers (*Calendula officinalis*, p. 73) that are a vivid yellow/orange color are likely to make good medicine. If they have been sitting on a shelf for 18 months, they will probably look drab and pale.

Buying Herbal Products

When buying products such as capsules, tablets, essential oils, pessaries, and tinctures, always check the label on the jar or packet. If it does not do the following, do not buy it:

- Name all constituents of the product
- State the recommended daily dosage
- State the weight of each capsule or tablet, or volume of bottle
- List the weight of each constituent of a capsule, tablet, etc.
- List the ratio of herb in the product (for example, 1:3, meaning 1 part herb to 3 parts liquid).